STEM LITTLE EXPLORERS · AUTUMN SCIENCE LAB

LAB PACK

Pumpkins & Potions

Scientist's name: _____



🧪 Colour-changing potion lab

Make red cabbage potion base, then test each ingredient. Colour the first circle with your guess, the second with what really happened.

Acid		Neutral		Base	

Ingredient	My guess	Result	Acid / neutral / base?
Lemon juice	○	○	A · N · B
Vinegar	○	○	A · N · B
Plain water	○	○	A · N · B
Sparkling water	○	○	A · N · B
Baking soda water	○	○	A · N · B
_____	○	○	A · N · B
_____	○	○	A · N · B

Which potion surprised you most? Why?

Lab rule: potions are for looking, not drinking. An adult stays in the lab.

🟢 Slime lab

Rate each slime you make: colour in 1 to 3 stars.

Slime	Stretchy	Gooey	Bouncy	My favourite?
Ghost slime (glue)	☆☆☆	☆☆☆	☆☆☆	☐
Monster snot (psyllium)	☆☆☆	☆☆☆	☆☆☆	☐
Frog-spawn potion (chia)	☆☆☆	☆☆☆	☆☆☆	☐
Bloody jelly goo (gelatine)	☆☆☆	☆☆☆	☆☆☆	☐
Mummy slime (marshmallow)	☆☆☆	☆☆☆	☆☆☆	☐
Pumpkin cloud slime	☆☆☆	☆☆☆	☆☆☆	☐

🔬 Activator experiment

Three batches of ghost slime. Change only the activator.

Activator	Stretch before it breaks (cm)	Does it bounce?	Seconds to flatten into a puddle
1 tablespoon			
2 tablespoons			
3 tablespoons			

🧟 Oobleck tests

Did the oobleck act like a solid or a liquid?

- **Punch it fast** ☐ Solid ☐ Liquid
- **Rest your hand on it** ☐ Solid ☐ Liquid
- **Roll it into a ball** ☐ Solid ☐ Liquid
- **Open your hand** ☐ Solid ☐ Liquid
- **Pull your hand out fast** ☐ Solid ☐ Liquid
- **Drop a toy on it** ☐ Solid ☐ Liquid

🎨 Draw your favourite slime

What colour is it? What would you call it?

Autumn colour hunt

Go outside and tick off everything you find. Look, but don't pick mushrooms!

- ☐ A red leaf
- ☐ A yellow leaf
- ☐ A chestnut or conker
- ☐ An acorn
- ☐ A pinecone
- ☐ A mushroom (just look)
- ☐ A spider web
- ☐ A pumpkin
- ☐ A feather
- ☐ An apple tree
- ☐ A snail
- ☐ A leaf with 3 colours

Spider rescue race

Time how long it takes to free one spider with each method.

Method	My guess (minutes)	Real time (minutes)
Salt only		
Warm water only		
Salt + warm water		

Which method won? Why do you think so?

Potion labels

Colour, cut out and stick on your jars. The blank ones are for your own inventions.



Dragon tears

NOT FOR DRINKING!



Bat wing juice

NOT FOR DRINKING!



Frog spawn

NOT FOR DRINKING!



Ghost goo

NOT FOR DRINKING!



Zombie goo

NOT FOR DRINKING!



Monster snot

NOT FOR DRINKING!



Midnight brew

NOT FOR DRINKING!



Swamp water

NOT FOR DRINKING!



Pumpkin elixir

NOT FOR DRINKING!



Witch's sneeze

NOT FOR DRINKING!

_____

NOT FOR DRINKING!

_____

NOT FOR DRINKING!